Multistrategy Improved Particle Swarm Optimization Algorithm for Path Planning of UAV in 3-D Low Altitude Urban Environment

(DA 243)

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Outline

1. Introduction
2. Build-up
3. Argument
4. Conclusion

Introduction

- Urban logistics is a key component of smart cities.
- But rapid urbanization and e-commerce growth have placed heavy pressure on delivery systems.
- Unmanned Aerial Vehicles (UAVs) offer an efficient alternative for last-mile logistics.
- However, autonomous UAV path planning in complex 3-D environments remains challenging.

- Heuristic methods like PSO and other swarm-based algorithms often suffer from slow convergence, high collision risk, and premature trapping in local optima.
- To address these limitations, this work proposes a **Multi-Strategy Improved Particle Swarm Optimization (MSIPSO)** algorithm.

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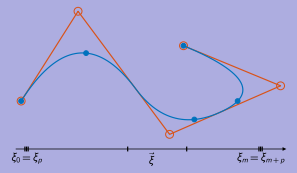
Building Up: Problem Formulation

Objective:

Find an optimal 3D path from start point S to target point T.

- **Path Representation (B-Splines):**
  - A B-spline is a continuous and twice differentiable smooth curve
  - 3 control points (P_1, P_2, P_3)
- **Optimization Variables:**
  - just 9 variables (x, y, z for three points)



Cost Function:

The total fitness function:

$$F = \lambda f_L + \mu f_S + \phi f_H + \gamma f_A$$

Where:

- **f_L**: Path Length Cost
- **f_s**: Path Safety (Collision) Cost
- **f_h**: Flight Altitude Cost
- **f_A**: Angle Change Cost
- **lambda, mu, phi, gamma**: Weighting coefficients determined by mission priorities

PSO

Pso algorithm is a type of swarm intelligence algorithm with wide applications in optimization problems. The mathematical expression for updating particle position and velocity is as follows:

$$p_i^{t+1} = p_i^t + v_i^{t+1}$$
$$v_i^{t+1} = wv_i^t + c_1 r_1 (s_i^t - p_i^t) + c_2 r_2 (g^t - p_i^t)$$

st_i: historical optimal position of individual
gt:historical optimal position of population
w : inertia weight parameter
c1, c2: learning factors
r1, r2: two random numbers in [0, 1]

The MSIPSO is built on top of the standard PSO algorithm.

Building Up: Continued...

Local Deadlock Escape Strategy (MSIPSO)

- In dense 3-D urban environments, standard PSO often becomes trapped in local deadlocks, where planned paths repeatedly collide with obstacles and the fitness remains excessively high. MSIPSO detects this condition using a fitness threshold $F_{\max} = K_{\max} \|p_{end} - p_{start}\|$
- If the global best satisfies **F(gt)>Fmax**, the algorithm enters deadlock escape mode. In this mode, velocity limits are removed and the traditional directional update is replaced by a randomized rule.

$$v_i^{t+1} = A - 2Ar_3,$$

$$A = \sqrt{\frac{L}{N}}$$

A: length of a certain dimension in the flight spac
N: is the number of particles
r3~U(0,1) = uniform random number in [0,1]

Diversified Evolutionary Startegy (MSIPSO)

- MSIPSO adopts a diversified evolutionary strategy by dividing particles into three subpopulations based on fitness: **D1** particles closest to the current global optimum, **D2** particles with moderate fitness, and **D3** particles far away from the global optimum.
- For particles in D1, slow, fine-grained local refinement is done:

$$p_i^{D_1}(t+1) = p_i^{D_1}(t) + \left(a_1 + (a_2 - a_1) \cdot \frac{t}{M}\right) \cdot v_i(t+1).$$

- For particles in D2, wider global exploration is done:

$$p_i^{D_2}(t+1) = p_i^{D_2}(t) + \left(b_1 + (b_2 - b_1) \cdot \frac{t}{M}\right) \cdot v_i(t+1).$$

$$a_1 < b_1,\ a_2 < b_2 \text{ and } (a_2 - a_1) < (b_2 - b_1)$$

- Particles in D3 are randomly updated to increase population diversity:

$$p_i^{D_3}(t+1) = r4 \cdot (p_{\max} - p_{\min})$$

r4~U(0,1) = uniform random number in [0,1]

Adaptive Non-linear parameter strategy:

In dense urban UAV path planning, the algorithm must:

- Early stage: Strong exploration
  - Search widely to quickly find feasible paths.
- Later stage: Strong exploitation
  - Converge precisely toward the optimal path.

Fixed PSO parameters cannot adapt to this changing requirement. As iterations progress, the balance between exploration and exploitation must shift dynamically.

MSIPSO introduces nonlinear adaptive parameters:

- Exponentially Decreasing Inertia Weight function:

$$w(t) = w_{\min} + (w_{\max} - w_{\min}) \cdot \frac{e^{-\frac{t}{T}} - e^{-1}}{e^{-\frac{1}{T}} - e^{-1}}$$

  - Starts large → encourages global exploration
  - Gradually decreases → strengthens local convergence
  - Nonlinear decay ensures smooth transition

- Logarithmic Asynchronous Learning Factors:

$$c_1(t) = c_{1max} - (c_{1max} - c_{1min}) \cdot \ln\left(1 + (e-1)\frac{t}{T}\right)$$

$$c_2(t) = c_{2min} + (c_{2max} - c_{2min}) \cdot \ln\left(1 + (e-1)\frac{t}{T}\right)$$

  - c_1(t) *(self-learning)* decreases over time
  - c_2(t) *(social learning)* increases over time

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WHY MSIPSO IS BETTER ?

1. Handles Dense Obstacles More Reliably

- Traditional PSO / GWO / WOA often get stuck in local deadlocks in complex 3-D environments.
- MSIPSO introduces a *local deadlock escape strategy*, allowing particles to jump out of infeasible regions and quickly find valid paths.
- Result: Fewer collisions and higher success rate in dense urban scenarios.

2. Faster and More Stable Convergence

- Fixed parameters in standard PSO cause either slow search or premature convergence.
- MSIPSO uses adaptive *nonlinear inertia weights* + *asynchronous learning factors* which allows early global exploration.

3. Better Exploration Using Multi-Population Strategy

- Most algorithms update all particles the same way.
- MSIPSO divides particles into high / medium / low quality groups, each with different update rules which greatly improves diversity and reduce chances of being stuck in local optima.

Result

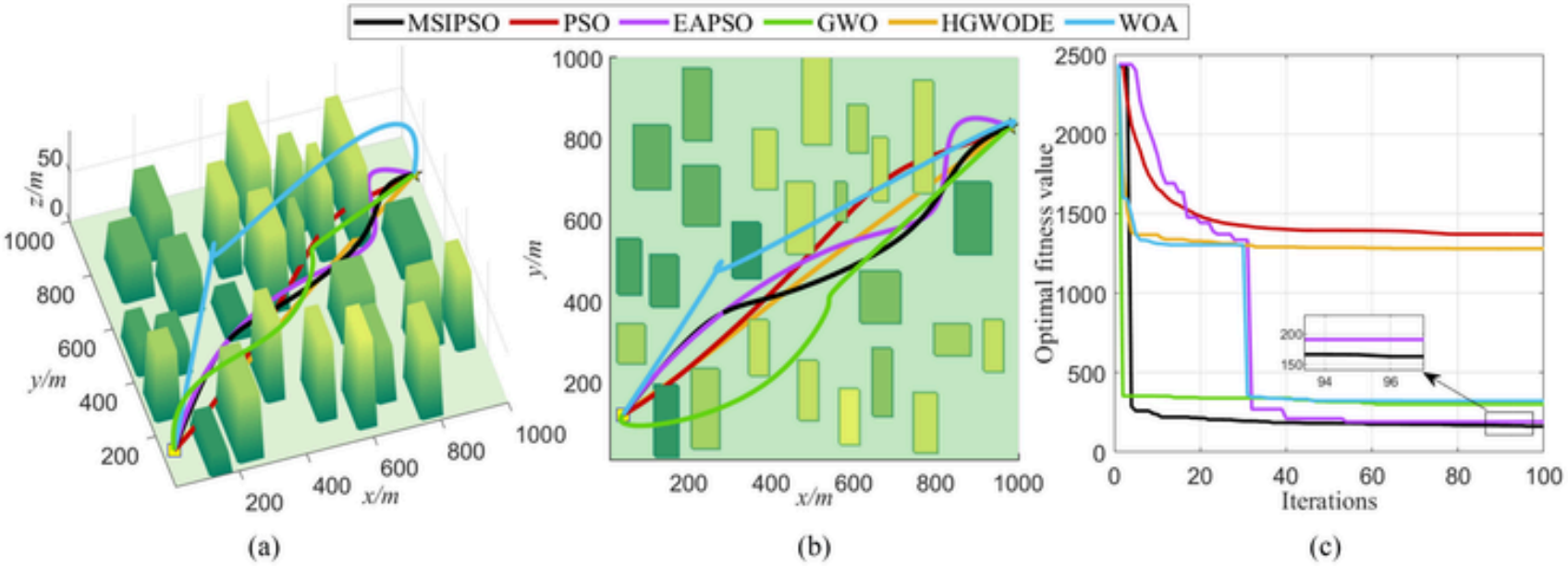

Fig. 10. Comparison of six algorithms in dense environments (scenario 1). (a) 3-D path. (b) Top view. (c) Optimal fitness with iteration.

TABLE III
COMPARISON OF STATISTICAL DATAS IN THE DENSE ENVIRONMENT

Algorithm	Maximum	Minimum	Average	Median	collision
MSIPSO	**144.8756**	**129.0051**	**137.9278**	**138.1727**	**2**
PSO	198.8338	138.0815	170.5604	170.1771	17
EAPSO	199.7229	141.8617	171.0855	172.3685	12
GWO	258.1692	170.5088	214.6596	214.5901	18
HGWODE	249.9329	144.5834	204.4240	207.2605	14
WOA	268.3170	234.5290	201.4433	236.9882	19

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Conclusion

- The proposed MSIPSO algorithm improves PSO by:
  - Introducing a local deadlock escape strategy
  - Using adaptive nonlinear inertia and learning factors
  - Applying multi-population differentiated evolution
- Simulation results demonstrate:
  - Faster convergence
  - Higher accuracy
  - Greater stability
  - Fewer collisions in dense environments
- Therefore, MSIPSO is a robust and effective solution for 3D UAV path planning in low-altitude urban scenarios

References

F. Tao, Z. Chen, Z. Wang, L. Zhu, and J. Wang, “Multistrategy Improved Particle Swarm Optimization Algorithm for Path Planning of UAV in 3-D Low Altitude Urban Environment,” IEEE Internet of Things Journal, vol. 12, no. 19, Oct. 2025.